

Course 5492

4 Credits: 4

Program Core

Source-grounded

# Microcontroller and

- To equip students on architecture, interfacing and Assembly Language

## OFFICIAL SOURCE DECLARATION

### How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5492 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 5492.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

## COURSE DASHBOARD

### Official course information

#### Course information

| Item                 | Official information                       |
|----------------------|--------------------------------------------|
| Programme            | Diploma in Microelectronics                |
| Course code          | 5492                                       |
| Course title         | Microcontroller and                        |
| Semester             | 4 Credits: 4                               |
| Course category      | Program Core                               |
| Periods per week     | 4 (L:3, T:1, P:0) Periods per semester: 60 |
| Periods per semester | 60                                         |

| Item         | Official information                                 |
|--------------|------------------------------------------------------|
| Credits      | 4                                                    |
| Prerequisite | See the official syllabus PDF                        |
| Assessment   | Use the official assessment section reproduced below |

## Modules

4 official module sections detected.

## Topic entries

16 source-aligned topic entries retained.

## Study mode

Theory and application emphasis.

### STUDY METHOD

## How to use this lesson

### First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

### Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

### Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

## Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

### LEARNING OUTCOMES

## Objectives, outcomes and mapping

### Course objectives

- To equip students on architecture, interfacing and Assembly Language
- Programming (ALP) of 8051 microcontrollers.
- To enable students to develop simple interfacing applications for daily use
- To prepare students for troubleshooting of microcontroller-based equipment.

### Course outcomes

| CO  | Official outcome                                                                | Study level           |
|-----|---------------------------------------------------------------------------------|-----------------------|
| CO1 | 12 Understanding Microcontroller. Develop simple Assembly Language Programs     | See official syllabus |
| CO2 | 17 Applying (ALP) for 8051. Explain interrupts, timer and serial communication  | See official syllabus |
| CO3 | 14 Understanding in 8051 Illustrate the interfacing of various I/O devices with | See official syllabus |
| CO4 | 15 Understanding                                                                | See official syllabus |

### CO-PO mapping evidence

| Row | Extracted official mapping |
|-----|----------------------------|
| CO1 | CO1 2                      |
| CO2 | CO2 3                      |
| CO3 | CO3 2                      |
| CO4 | CO4 2                      |

### COVERAGE AUDIT

# Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

## Module coverage

| Module   | Official heading                                                | Topic entries | Hours        |
|----------|-----------------------------------------------------------------|---------------|--------------|
| Module 1 | Explain the features and block diagram of 8051 Microcontroller. | 4             | As specified |
| Module 2 | Develop simple Assembly Language Programs (ALP) for 8051.       | 4             | As specified |
| Module 3 | Explain interrupts, timer and serial communication in 8051.     | 4             | As specified |
| Module 4 | Illustrate the interfacing of various I/O devices with 8051     | 4             | As specified |

## MODULE 1

# Explain the features and block diagram of 8051 Microcontroller.

This module is presented from the official syllabus scope for Microcontroller and. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Explain the features and block diagram of 8051 Microcontroller.
- Official duration: As specified

- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## Compare Microprocessor and Microcontroller

Compare Microprocessor and Microcontroller is an official topic in Module 1 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Compare Microprocessor and Microcontroller using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, compare microprocessor and microcontroller should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                      |
|---------------------|--------------------------------------------------------------------------------------|
| Definition/identity | What Compare Microprocessor and Microcontroller means in the official course context |
| Study lens          | Principle → components or stages → result → application                              |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate    |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-Compare Microprocessor and Microcontroller  
Microprocessor definition/meaning. Microprocessor steps, application. Technical terms English-Malayalam.

## List the features of 8051 microcontroller. 1 Remembering Explain the block diagram of 8051

List the features of 8051 microcontroller. 1 Remembering Explain the block diagram of 8051 is an official topic in Module 1 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify List the features of 8051 microcontroller. 1 Remembering Explain the block diagram of 8051 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, list the features of 8051 microcontroller. 1 remembering explain the block diagram of 8051 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                      |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What List the features of 8051 microcontroller. 1 Remembering Explain the block diagram of 8051 means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                              |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                    |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.



**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-  
List the features of 8051 microcontroller. 1 Remembering Explain the block diagram of 8051   
definition/meaning . . . . . , steps, application   
 . . . . . . Technical terms English-  
 . . . . .

## 2 Understanding microcontroller.

2 Understanding microcontroller. is an official topic in Module 1 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding microcontroller. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding microcontroller. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What 2 Understanding microcontroller. means in the official course context        |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-2 Understanding microcontroller.   
 definition/meaning .   
 application   
 Technical terms English-   
 .

**Explain the memory of 8051. 7 Understanding Features and Block diagram of 8051 Introduction to microprocessors and microcontrollers, features of 8051, simple block diagram of 8051, Data memory organization, program memory organization, PSW register.**

Explain the memory of 8051. 7 Understanding Features and Block diagram of 8051 Introduction to microprocessors and microcontrollers, features of 8051, simple block diagram of 8051, Data memory organization, program memory organization, PSW register. is an official topic in Module 1 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain the memory of 8051. 7 Understanding Features and Block diagram of 8051 Introduction to microprocessors and microcontrollers, features of 8051, simple block diagram of 8051, Data memory organization, program memory organization, PSW register. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain the memory of 8051. 7 understanding features and block diagram of 8051 introduction to microprocessors and microcontrollers, features of 8051, simple block diagram of 8051, data memory organization, program memory organization, psw register. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Explain the memory of 8051. 7 Understanding Features and Block diagram of 8051 Introduction to microprocessors and microcontrollers, features of 8051, simple block diagram of |

| Aspect        | What to prepare                                                                                                 |
|---------------|-----------------------------------------------------------------------------------------------------------------|
|               | 8051, Data memory organization, program memory organization, PSW register. means in the official course context |
| Study lens    | Principle → components or stages → result → application                                                         |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate                               |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1- Explain the memory of 8051. 7 Understanding Features and Block diagram of 8051 Introduction to microprocessors and microcontrollers, features of 8051, simple block diagram of 8051, Data memory organization, program memory organization, PSW register. ഓരോ ഓരോന്നിനും നിർവ്വചനം/അർത്ഥം നൽകുക. ഓരോന്നിനും ഓരോന്നിനും, steps, application നൽകുക. ഓരോന്നിനും ഓരോന്നിനും. Technical terms English-ൽ നൽകുക. ഓരോന്നിനും ഓരോന്നിനും.

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.  
Educational explanations below are separate elaboration.

Explain the features and block diagram of 8051 Microcontroller.

|       |                                                          |   |
|-------|----------------------------------------------------------|---|
| M1.01 | Compare Microprocessor and Microcontroller Understanding | 2 |
|-------|----------------------------------------------------------|---|

|       |                                                        |   |
|-------|--------------------------------------------------------|---|
| M1.02 | List the features of 8051 microcontroller. Remembering | 1 |
|-------|--------------------------------------------------------|---|

|       |                                                                  |   |
|-------|------------------------------------------------------------------|---|
| M1.03 | Explain the block diagram of 8051 microcontroller. Understanding | 2 |
|-------|------------------------------------------------------------------|---|

|       |                                           |   |
|-------|-------------------------------------------|---|
| M1.04 | Explain the memory of 8051. Understanding | 7 |
|-------|-------------------------------------------|---|

Contents:

Features and Block diagram of 8051

Introduction to microprocessors and microcontrollers, features of 8051, simple block

diagram of 8051, Data memory organization, program memory organization, PSW register.

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

## MODULE 2

# Develop simple Assembly Language Programs (ALP) for 8051.

This module is presented from the official syllabus scope for Microcontroller and. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Develop simple Assembly Language Programs (ALP) for 8051.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

### 3 Understanding Programming. Explain various addressing modes and types

3 Understanding Programming. Explain various addressing modes and types is an official topic in Module 2 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### Study points

- Define or identify 3 Understanding Programming. Explain various addressing modes and types using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### Engineering or professional relevance

In Polytechnic practice, 3 understanding programming. explain various addressing modes and types should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 3 Understanding Programming. Explain various addressing modes and types means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                 |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related



but different term.

**Malayalam support:** Module 2- 3 Understanding Programming. Explain various addressing modes and types  $\text{പ്രതിബാധനാ രീതികൾ}$  definition/meaning  $\text{പ്രതിബാധനാ രീതികൾ}$ .  $\text{പ്രതിബാധനാ രീതികൾ}$   $\text{പ്രതിബാധനാ രീതികൾ}$ , steps, application  $\text{പ്രതിബാധനാ രീതികൾ}$   $\text{പ്രതിബാധനാ രീതികൾ}$ . Technical terms English-  $\text{പ്രതിബാധനാ രീതികൾ}$ .

## 4 Understanding of instructions. Describe Jump, Loop, Call and single bit level

4 Understanding of instructions. Describe Jump, Loop, Call and single bit level is an official topic in Module 2 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding of instructions. Describe Jump, Loop, Call and single bit level using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding of instructions. describe jump, loop, call and single bit level should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding of instructions. Describe Jump, Loop, Call and single bit level means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                   |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                         |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 2-4 Understanding of instructions. Describe Jump, Loop, Call and single bit level instructions definition/meaning and application. steps, application of instructions. Technical terms English-4 instructions.

## 4 Understanding instructions. Develop Assembly language programs for

4 Understanding instructions. Develop Assembly language programs for is an official topic in Module 2 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding instructions. Develop Assembly language programs for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding instructions. develop assembly language programs for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                |
|---------------------|----------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding instructions. Develop Assembly language programs for means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                        |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                              |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 2-4 Understanding instructions. Develop Assembly language programs for  $\mu\text{PIC16F877}$  definition/meaning  $\mu\text{PIC16F877}$ .  $\mu\text{PIC16F877}$   $\mu\text{PIC16F877}$ , steps, application  $\mu\text{PIC16F877}$   $\mu\text{PIC16F877}$ . Technical terms English- $\mu\text{PIC16F877}$   $\mu\text{PIC16F877}$ .

**6 Applying 8051. Introduction to Assembly Language programming of 8051 – Structure, addressing modes, Types of instructions: -Data transfer, arithmetic, logical, Compare, Rotate and swap instructions. Describe jump instruction-conditional and unconditional, call instructions, calling subroutines. machine cycle, Delay generation. Single-Bit level instructions, Data serialization. Simple programs: - 8-bit Addition, 8-bit subtraction, multiplication, division, ASCII to packed BCD, packed BCD to ASCII, BCD to hex and hex to BCD, port reading and writing.**

6 Applying 8051. Introduction to Assembly Language programming of 8051 – Structure, addressing modes, Types of instructions: -Data transfer, arithmetic, logical, Compare, Rotate and swap instructions. Describe jump instruction-conditional and unconditional, call instructions, calling subroutines. machine cycle, Delay generation. Single-Bit level instructions, Data serialization. Simple programs: - 8-bit Addition, 8-bit subtraction, multiplication, division, ASCII to packed BCD, packed BCD to ASCII, BCD to hex and hex to BCD, port reading and writing. is an official topic in Module 2 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify 6 Applying 8051. Introduction to Assembly Language programming of 8051 – Structure, addressing modes, Types of instructions: -Data transfer, arithmetic, logical, Compare, Rotate and swap instructions. Describe jump instruction- conditional and unconditional, call instructions, calling subroutines. machine cycle, Delay generation. Single-Bit level instructions, Data serialization. Simple programs: - 8-bit Addition, 8-bit subtraction, multiplication, division, ASCII to packed BCD, packed BCD to ASCII, BCD to hex and hex to BCD, port reading and writing. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, 6 applying 8051. introduction to assembly language programming of 8051 – structure, addressing modes, types of instructions: -data transfer, arithmetic, logical, compare, rotate and swap instructions. describe jump instruction- conditional and unconditional, call instructions, calling subroutines. machine cycle, delay generation. single-bit level instructions, data serialization. simple programs: - 8-bit addition, 8-bit subtraction, multiplication, division, ascii to packed bcd, packed bcd to ascii, bcd to hex and hex to bcd, port reading and writing. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 6 Applying 8051. Introduction to Assembly Language programming of 8051 – Structure, addressing modes, Types of instructions: -Data transfer, arithmetic, logical, Compare, Rotate and swap instructions. Describe jump instruction- conditional and unconditional, call instructions, calling subroutines. machine cycle, Delay generation. Single-Bit level instructions, Data serialization. Simple programs: - 8-bit Addition, 8-bit subtraction, multiplication, division, ASCII to packed BCD, packed BCD to ASCII, BCD to hex and hex to BCD, port reading and writing. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- 6 Applying 8051. Introduction to Assembly Language programming of 8051 – Structure, addressing modes, Types of instructions: -Data transfer, arithmetic, logical, Compare, Rotate and swap instructions. Describe jump instruction- conditional and unconditional, call instructions, calling subroutines. machine cycle, Delay generation. Single-Bit level instructions, Data serialization. Simple programs: - 8-bit Addition, 8-bit subtraction, multiplication, division, ASCII to packed BCD, packed BCD to ASCII, BCD to hex and hex to BCD, port reading and writing. definition/meaning. steps, application. Technical terms English-

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Develop simple Assembly Language Programs (ALP) for 8051.

Explain the structure of Assembly Language

M2.01 Understanding 3

Programming.

Explain various addressing modes and types

M2.02 Understanding 4

of instructions.

Describe Jump, Loop, Call and single bit level

M2.03 Understanding 4

instructions.

Develop Assembly language programs for

M2.04 Applying 6

8051.

Series Test – I 1

Contents:

Introduction to Assembly Language programming of 8051 – Structure, addressing modes, Types of instructions: -Data transfer, arithmetic, logical, Compare, Rotate and swap

instructions. Describe jump instruction- conditional and unconditional, call instructions,

calling subroutines. machine cycle, Delay generation. Single-Bit level instructions, Data

serialization. Simple programs: - 8-bit Addition, 8-bit subtraction,

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

### MODULE 3

## Explain interrupts, timer and serial communication in 8051.

This module is presented from the official syllabus scope for Microcontroller and. Use the topic cards for learning and the source transcript for exact coverage.



**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Explain interrupts, timer and serial communication in 8051.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## 4 Understanding Interrupts and handling of interrupts. Explain TMOD register and timer operating

4 Understanding Interrupts and handling of interrupts. Explain TMOD register and timer operating is an official topic in Module 3 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding Interrupts and handling of interrupts. Explain TMOD register and timer operating using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding interrupts and handling of interrupts. explain tmod register and timer operating should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding Interrupts and handling of interrupts. Explain TMOD register and timer operating means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                    |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                          |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-4 Understanding Interrupts and handling of interrupts. Explain TMOD register and timer operating definition/meaning. steps, application. Technical terms English-

## 4 Understanding modes. Illustrate TCON register and time delay

4 Understanding modes. Illustrate TCON register and time delay is an official topic in Module 3 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding modes. Illustrate TCON register and time delay using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding modes. illustrate tcon register and time delay should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                          |
|---------------------|----------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding modes. Illustrate TCON register and time delay means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                  |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                        |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 3-4 Understanding modes. Illustrate TCON register and time delay തിരഞ്ഞെടുക്കൽ നിർവ്വചന/അർത്ഥം തിരഞ്ഞെടുക്കൽ. തിരഞ്ഞെടുക്കൽ തിരഞ്ഞെടുക്കൽ, steps, application തിരഞ്ഞെടുക്കൽ തിരഞ്ഞെടുക്കൽ തിരഞ്ഞെടുക്കൽ. Technical terms English- തിരഞ്ഞെടുക്കൽ തിരഞ്ഞെടുക്കൽ.

### 3 Understanding calculation for Timer. Illustrate special function registers for serial

3 Understanding calculation for Timer. Illustrate special function registers for serial is an official topic in Module 3 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### Study points

- Define or identify 3 Understanding calculation for Timer. Illustrate special function registers for serial using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### Engineering or professional relevance

In Polytechnic practice, 3 understanding calculation for timer. illustrate special function registers for serial should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 3 Understanding calculation for Timer. Illustrate special function registers for serial means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                 |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-3 Understanding calculation for Timer. Illustrate special function registers for serial  $\text{SPI}$  definition/meaning  $\text{SPI}$ .  $\text{SPI}$   $\text{SPI}$   $\text{SPI}$ , steps, application  $\text{SPI}$   $\text{SPI}$ . Technical terms English- $\text{SPI}$   $\text{SPI}$   $\text{SPI}$ .

**3 Understanding communication. Interrupts, Timers, Serial communication. 8051 Interrupts: Concept, Types, Interrupt service routine (ISR), IE register, Steps in executing an interrupt, Interrupt priority, IP register. Timer: TMOD register, Timer operating modes, TCON register, Time Delay calculation. Basics of serial communication, Baud rate, 8051 connections to RS232 connector, SBUF register, SCON register and Different serial data transmission modes in 8051. PCON register.**

3 Understanding communication. Interrupts, Timers, Serial communication. 8051 Interrupts: Concept, Types, Interrupt service routine (ISR), IE register, Steps in executing an interrupt, Interrupt priority, IP register. Timer: TMOD register, Timer operating modes, TCON register, Time Delay calculation. Basics of serial communication, Baud rate, 8051 connections to RS232 connector, SBUF register, SCON register and Different serial data transmission modes in 8051. PCON register. is an official topic in Module 3 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 3 Understanding communication. Interrupts, Timers, Serial communication. 8051 Interrupts: Concept, Types, Interrupt service routine (ISR), IE register, Steps in executing an interrupt, Interrupt priority, IP register. Timer: TMOD register, Timer operating modes, TCON register, Time Delay calculation. Basics of serial communication, Baud rate, 8051 connections to RS232 connector, SBUF register, SCON register and Different serial data transmission modes in 8051. PCON register. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 3 understanding communication. interrupts, timers, serial communication. 8051 interrupts: concept, types, interrupt service routine (isr), ie



register, steps in executing an interrupt, interrupt priority, ip register. timer: tmod register, timer operating modes, tcon register, time delay calculation. basics of serial communication, baud rate, 8051 connections to rs232 connector, sbuf register, scon register and different serial data transmission modes in 8051. pcon register. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 3 Understanding communication. Interrupts, Timers, Serial communication. 8051 Interrupts: Concept, Types, Interrupt service routine (ISR), IE register, Steps in executing an interrupt, Interrupt priority, IP register. Timer: TMOD register, Timer operating modes, TCON register, Time Delay calculation. Basics of serial communication, Baud rate, 8051 connections to RS232 connector, SBUF register, SCON register and Different serial data transmission modes in 8051. PCON register. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- 3 Understanding communication. Interrupts, Timers, Serial communication. 8051 Interrupts: Concept, Types, Interrupt service routine (ISR), IE register, Steps in executing an interrupt, Interrupt priority, IP register. Timer: TMOD register, Timer operating modes, TCON register, Time Delay calculation. Basics of serial communication, Baud rate, 8051 connections to RS232 connector, SBUF register, SCON register and Different serial data transmission modes in 8051. PCON register. definition/meaning steps, application . Technical terms English- .

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Explain interrupts, timer and serial communication in 8051.

|                                                                                         |                                                  |   |
|-----------------------------------------------------------------------------------------|--------------------------------------------------|---|
| M3.01                                                                                   | Explain concept, types, priority, Registers for  | 4 |
| Understanding                                                                           | Interrupts and handling of interrupts.           |   |
| M3.02                                                                                   | Explain TMOD register and timer operating        | 4 |
| Understanding                                                                           | modes.                                           |   |
| M3.03                                                                                   | Illustrate TCON register and time delay          | 3 |
| Understanding                                                                           | calculation for Timer.                           |   |
| M3.04                                                                                   | Illustrate special function registers for serial | 3 |
| Understanding                                                                           | communication.                                   |   |
| Contents:                                                                               |                                                  |   |
| Interrupts, Timers, Serial communication.                                               |                                                  |   |
| 8051 Interrupts: Concept, Types, Interrupt service routine (ISR), IE register, Steps in |                                                  |   |
| executing an interrupt, Interrupt priority, IP register.                                |                                                  |   |
| Timer: TMOD register, Timer operating modes, TCON register, Time Delay                  |                                                  |   |
| calculation.                                                                            |                                                  |   |
| Basics of serial communication, Baud rate, 8051 connections to RS232 connector, SBUF    |                                                  |   |

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

### MODULE 4

## Illustrate the interfacing of various I/O devices with 8051

This module is presented from the official syllabus scope for Microcontroller and. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Illustrate the interfacing of various I/O devices with 8051
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## 4 Understanding 8051. Explain the interfacing of DC motor for speed

4 Understanding 8051. Explain the interfacing of DC motor for speed is an official topic in Module 4 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding 8051. Explain the interfacing of DC motor for speed using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding 8051. explain the interfacing of dc motor for speed should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding 8051. Explain the interfacing of DC motor for speed means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                       |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                             |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 4- 4 Understanding 8051. Explain the interfacing of DC motor for speed control. Give definition/meaning of speed control. Give steps, application of speed control. Technical terms English- Malayalam.

### 3 Understanding control. Illustrate the interfacing of 4x4 keyboard and

3 Understanding control. Illustrate the interfacing of 4x4 keyboard and is an official topic in Module 4 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### Study points

- Define or identify 3 Understanding control. Illustrate the interfacing of 4x4 keyboard and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### Engineering or professional relevance

In Polytechnic practice, 3 understanding control. illustrate the interfacing of 4x4 keyboard and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 3 Understanding control. Illustrate the interfacing of 4x4 keyboard and means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                 |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 4-3 Understanding control. Illustrate the interfacing of 4x4 keyboard and  $\mu$ controller definition/meaning  $\mu$ controller  $\mu$ controller  $\mu$ controller, steps, application  $\mu$ controller  $\mu$ controller. Technical terms English- $\mu$  controller  $\mu$ controller.

## 4 Understanding 16x2 LCD system with 8051. Illustrate ADC, DAC and LM35 temperature

4 Understanding 16x2 LCD system with 8051. Illustrate ADC, DAC and LM35 temperature is an official topic in Module 4 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding 16x2 LCD system with 8051. Illustrate ADC, DAC and LM35 temperature using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding 16x2 lcd system with 8051. illustrate adc, dac and lm35 temperature should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                               |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding 16x2 LCD system with 8051. Illustrate ADC, DAC and LM35 temperature means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                       |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                             |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.



**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- 4 Understanding 16x2 LCD system with 8051. Illustrate ADC, DAC and LM35 temperature sensor definition/meaning and applications. Explain operation, steps, application of ADC and DAC. Technical terms English- Malayalam.

**4 Understanding sensor interfacing with 8051. Stepper motor - operation, explain the interfacing with 8051 DC motor - operation, explain the interfacing with 8051 4x4 -Matrix keyboard - Structure, identifying key press, explain the interfacing with 8051. 16x2 LCD - Pin description, advantages of LCD, explain the interfacing with 8051. Explain the interfacing of 8051 - with ADC, with DAC, with temperature control system using LM35 temperature sensor. Text / Reference: T/R Book Title/Author Muhammad Ali Mazidi and Janice Gillispie Mazidi - The 8051 Microcontroller T1 and Embedded Systems Using Assembly and C - Pearson Education- Second Edition. Subratha Ghoshal - 8051 microcontroller internals, instructions, programming T2 and interfacing- Pearson T3 Kenneth J Ayala - The 8051 Microcontroller - Thomson - Third Edition. Online Resources: Sl.No Website Link 1 <https://www.electronicshub.org> 2 <https://www.elprocus.com> 3 <https://www.tutorialspoint.com>**

4 Understanding sensor interfacing with 8051. Stepper motor - operation, explain the interfacing with 8051 DC motor - operation, explain the interfacing with 8051 4x4 -Matrix keyboard - Structure, identifying key press, explain the interfacing with 8051. 16x2 LCD - Pin description, advantages of LCD, explain the interfacing with 8051. Explain the interfacing of 8051 - with ADC, with DAC, with temperature control system using LM35 temperature sensor. Text / Reference: T/R Book Title/Author Muhammad Ali Mazidi and Janice Gillispie Mazidi - The 8051 Microcontroller T1 and Embedded Systems Using Assembly and C - Pearson Education- Second Edition. Subratha Ghoshal - 8051 microcontroller internals, instructions, programming T2 and interfacing- Pearson T3 Kenneth J Ayala - The 8051 Microcontroller - Thomson - Third Edition. Online Resources: Sl.No Website Link 1 <https://www.electronicshub.org> 2 <https://www.elprocus.com> 3 <https://www.tutorialspoint.com> is an official topic in Module 4 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify 4 Understanding sensor interfacing with 8051. Stepper motor - operation, explain the interfacing with 8051 DC motor - operation, explain the interfacing with 8051 4x4 -Matrix keyboard - Structure, identifying key press,

explain the interfacing with 8051. 16x2 LCD - Pin description, advantages of LCD, explain the interfacing with 8051. Explain the interfacing of 8051 - with ADC, with DAC, with temperature control system using LM35 temperature sensor. Text / Reference: T/R Book Title/Author Muhammad Ali Mazidi and Janice Gillispie Mazidi - The 8051 Microcontroller T1 and Embedded Systems Using Assembly and C - Pearson Education- Second Edition. Subratha Ghoshal – 8051 microcontroller internals, instructions, programming T2 and interfacing- Pearson T3 Kenneth J Ayala - The 8051 Microcontroller - Thomson - Third Edition. Online Resources: Sl.No Website Link 1 <https://www.electronicshub.org> 2 <https://www.elprocus.com> 3 <https://www.tutorialspoint.com> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, 4 understanding sensor interfacing with 8051. stepper motor – operation, explain the interfacing with 8051 dc motor – operation, explain the interfacing with 8051 4x4 -matrix keyboard – structure, identifying key press, explain the interfacing with 8051. 16x2 lcd - pin description, advantages of lcd, explain the interfacing with 8051. explain the interfacing of 8051 - with adc, with dac, with temperature control system using lm35 temperature sensor. text / reference: t/r book title/author muhammad ali mazidi and janice gillispie mazidi - the 8051 microcontroller t1 and embedded systems using assembly and c - pearson education- second edition. subratha ghoshal – 8051 microcontroller internals, instructions, programming t2 and interfacing- pearson t3 kenneth j ayala - the 8051 microcontroller - thomson - third edition. online resources: sl.no website link 1 <https://www.electronicshub.org> 2 <https://www.elprocus.com> 3 <https://www.tutorialspoint.com> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding sensor interfacing with 8051. Stepper motor – operation, explain the interfacing with 8051 DC motor – operation, explain the interfacing with 8051 4x4 -Matrix keyboard – Structure, identifying key press, explain the interfacing with 8051. 16x2 LCD - Pin description, advantages of LCD, explain the interfacing with 8051. Explain the interfacing of 8051 - with ADC, with DAC, with temperature control system using LM35 temperature sensor. Text / Reference: T/R Book Title/Author Muhammad Ali Mazidi and Janice Gillispie Mazidi - The 8051 Microcontroller T1 and Embedded Systems Using Assembly and C - Pearson Education- Second Edition. Subratha Ghoshal – 8051 microcontroller internals, instructions, programming T2 and interfacing- Pearson T3 Kenneth J Ayala - The 8051 Microcontroller - Thomson - Third Edition. Online Resources: Sl.No Website Link 1 <a href="https://www.electronicshub.org">https://www.electronicshub.org</a> 2 <a href="https://www.elprocus.com">https://www.elprocus.com</a> 3 <a href="https://www.tutorialspoint.com">https://www.tutorialspoint.com</a> means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- 4 Understanding sensor interfacing with 8051. Stepper motor – operation, explain the interfacing with 8051 DC motor – operation, explain the interfacing with 8051 4x4 -Matrix keyboard – Structure, identifying key press, explain the interfacing with 8051. 16x2 LCD - Pin description, advantages of LCD, explain the interfacing with 8051. Explain the interfacing of 8051 - with ADC, with DAC, with temperature control system using LM35 temperature sensor. Text / Reference: T/R Book Title/Author Muhammad Ali Mazidi and Janice Gillispie Mazidi - The 8051 Microcontroller T1 and Embedded Systems Using Assembly and C - Pearson Education- Second Edition. Subratha Ghoshal – 8051 microcontroller internals, instructions, programming T2 and interfacing- Pearson T3 Kenneth J Ayala - The 8051 Microcontroller - Thomson - Third Edition. Online Resources: Sl.No Website Link 1 <https://www.electronicshub.org> 2 <https://www.elprocus.com> 3 <https://www.tutorialspoint.com> ഏകദേശം definition/meaning ഏകദേശം. ഏകദേശം ഏകദേശം, steps, application ഏകദേശം. Technical terms English-ഏകദേശം.

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Illustrate the interfacing of various I/O devices with 8051

|               |                                                                                             |   |
|---------------|---------------------------------------------------------------------------------------------|---|
| M4.01         | Explain the interfacing of stepper motor with 8051.                                         | 4 |
| Understanding |                                                                                             |   |
| M4.02         | Explain the interfacing of DC motor for speed control.                                      | 3 |
| Understanding |                                                                                             |   |
| M4.03         | Illustrate the interfacing of 4x4 keyboard and 16x2 LCD system with 8051.                   | 4 |
| Understanding |                                                                                             |   |
| M4.04         | Illustrate ADC, DAC and LM35 temperature sensor interfacing with 8051.                      | 4 |
| Understanding |                                                                                             |   |
|               | Series Test – II                                                                            | 1 |
| Contents:     |                                                                                             |   |
|               | Stepper motor – operation, explain the interfacing with 8051                                |   |
|               | DC motor – operation, explain the interfacing with 8051                                     |   |
|               | 4x4 -Matrix keyboard – Structure, identifying key press, explain the interfacing with 8051. |   |
|               | 16x2 LCD - Pin description, advantages of LCD, explain the interfacing with 8051.           |   |
|               | Explain the interfacing of 8051 - with ADC, with DAC, with temperature control              |   |

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

### RAPID REFERENCE

## Concept and formula bank

### Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

### Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

## RAPID REVISION

### Diagram and source-transcript library

#### Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

#### Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

#### Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

#### Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

## OFFICIAL SELF-LEARNING

# Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

## EXAMINATION PREPARATION

### Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

## ACTIVE RECALL

### Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

#### Module 1

**Define or explain Compare Microprocessor and Microcontroller. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Compare Microprocessor and Microcontroller*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain List the features of 8051 microcontroller. 1 Remembering Explain the block diagram of 8051. (3 marks)**

**Answer:** Begin with the standard definition or identity of *List the features of 8051 microcontroller. 1 Remembering Explain the block diagram of 8051*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 2 Understanding microcontroller.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *2 Understanding microcontroller..*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Explain the memory of 8051. 7 Understanding Features and Block diagram of 8051 Introduction to microprocessors and microcontrollers, features of 8051, simple block diagram of 8051, Data memory organization, program memory organization, PSW register.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Explain the memory of 8051. 7 Understanding Features and Block diagram of 8051 Introduction to microprocessors and microcontrollers, features of 8051, simple block diagram of 8051, Data memory organization, program memory organization, PSW register..*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.



**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

## Module 2

**Define or explain 3 Understanding Programming. Explain various addressing modes and types. (3 marks)**

**Answer:** Begin with the standard definition or identity of *3 Understanding Programming*. Explain various addressing modes and types. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 4 Understanding of instructions. Describe Jump, Loop, Call and single bit level. (3 marks)**

**Answer:** Begin with the standard definition or identity of *4 Understanding of instructions*. Describe Jump, Loop, Call and single bit level. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 4 Understanding instructions. Develop Assembly language programs for. (3 marks)**

**Answer:** Begin with the standard definition or identity of *4 Understanding instructions*. Develop Assembly language programs for. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps,

□□□□□ application □□□□ □□□□□□□□ □□□□□.

**Define or explain 6 Applying 8051. Introduction to Assembly Language programming of 8051 - Structure, addressing modes, Types of instructions: - Data transfer, arithmetic, logical, Compare, Rotate and swap instructions. Describe jump instruction- conditional and unconditional, call instructions, calling subroutines. machine cycle, Delay generation. Single-Bit level instructions, Data serialization. Simple programs: - 8-bit Addition, 8-bit subtraction, multiplication, division, ASCII to packed BCD, packed BCD to ASCII, BCD to hex and hex to BCD, port reading and writing.. (3 marks)**

**Answer:** Begin with the standard definition or identity of 6 Applying 8051. Introduction to Assembly Language programming of 8051 - Structure, addressing modes, Types of instructions: -Data transfer, arithmetic, logical, Compare, Rotate and swap instructions. Describe jump instruction- conditional and unconditional, call instructions, calling subroutines. machine cycle, Delay generation. Single-Bit level instructions, Data serialization. Simple programs: - 8-bit Addition, 8-bit subtraction, multiplication, division, ASCII to packed BCD, packed BCD to ASCII, BCD to hex and hex to BCD, port reading and writing.. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□ application □□□□ □□□□□□□□ □□□□□.

## Module 3

**Define or explain 4 Understanding Interrupts and handling of interrupts. Explain TMOD register and timer operating. (3 marks)**

**Answer:** Begin with the standard definition or identity of 4 Understanding Interrupts and handling of interrupts. Explain TMOD register and timer operating. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□ application □□□□ □□□□□□□□ □□□□□.

**Define or explain 4 Understanding modes. Illustrate TCON register and time delay. (3 marks)**

**Answer:** Begin with the standard definition or identity of *4 Understanding modes*. *Illustrate TCON register and time delay*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 3 Understanding calculation for Timer. Illustrate special function registers for serial. (3 marks)**

**Answer:** Begin with the standard definition or identity of *3 Understanding calculation for Timer*. *Illustrate special function registers for serial*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 3 Understanding communication. Interrupts, Timers, Serial communication. 8051 Interrupts: Concept, Types, Interrupt service routine (ISR), IE register, Steps in executing an interrupt, Interrupt priority, IP register. Timer: TMOD register, Timer operating modes, TCON register, Time Delay calculation. Basics of serial communication, Baud rate, 8051 connections to RS232 connector, SBUF register, SCON register and Different serial data transmission modes in 8051. PCON register.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *3 Understanding communication. Interrupts, Timers, Serial communication. 8051 Interrupts: Concept, Types, Interrupt service routine (ISR), IE register, Steps in executing an interrupt, Interrupt priority, IP register. Timer: TMOD register, Timer operating modes, TCON register, Time Delay calculation. Basics of serial communication, Baud rate, 8051 connections to RS232 connector, SBUF register, SCON register and Different serial data transmission modes in 8051. PCON register..* State the governing principle, list the key

components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

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## Module 4

**Define or explain 4 Understanding 8051. Explain the interfacing of DC motor for speed. (3 marks)**

**Answer:** Begin with the standard definition or identity of *4 Understanding 8051*. Explain the interfacing of DC motor for speed. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 3 Understanding control. Illustrate the interfacing of 4x4 keyboard and. (3 marks)**

**Answer:** Begin with the standard definition or identity of *3 Understanding control*. Illustrate the interfacing of 4x4 keyboard and. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 4 Understanding 16x2 LCD system with 8051. Illustrate ADC, DAC and LM35 temperature. (3 marks)**

**Answer:** Begin with the standard definition or identity of *4 Understanding 16x2 LCD system with 8051*. Illustrate ADC, DAC and LM35 temperature. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

**Define or explain 4 Understanding sensor interfacing with 8051. Stepper motor - operation, explain the interfacing with 8051 DC motor - operation, explain the interfacing with 8051 4x4 -Matrix keyboard - Structure, identifying key press, explain the interfacing with 8051. 16x2 LCD - Pin description, advantages of LCD, explain the interfacing with 8051. Explain the interfacing of 8051 - with ADC, with DAC, with temperature control system using LM35 temperature sensor. Text / Reference: T/R Book Title/Author Muhammad Ali Mazidi and Janice Gillispie Mazidi - The 8051 Microcontroller T1 and Embedded Systems Using Assembly and C - Pearson Education- Second Edition. Subratha Ghoshal - 8051 microcontroller internals, instructions, programming T2 and interfacing- Pearson T3 Kenneth J Ayala - The 8051 Microcontroller - Thomson - Third Edition. Online Resources: Sl.No Website Link 1 <https://www.electronicshub.org> 2 <https://www.elprocus.com> 3 <https://www.tutorialspoint.com>. (3 marks)**

**Answer:** Begin with the standard definition or identity of 4 Understanding sensor interfacing with 8051. Stepper motor - operation, explain the interfacing with 8051 DC motor - operation, explain the interfacing with 8051 4x4 -Matrix keyboard - Structure, identifying key press, explain the interfacing with 8051. 16x2 LCD - Pin description, advantages of LCD, explain the interfacing with 8051. Explain the interfacing of 8051 - with ADC, with DAC, with temperature control system using LM35 temperature sensor. Text / Reference: T/R Book Title/Author Muhammad Ali Mazidi and Janice Gillispie Mazidi - The 8051 Microcontroller T1 and Embedded Systems Using Assembly and C - Pearson Education- Second Edition. Subratha Ghoshal - 8051 microcontroller internals, instructions, programming T2 and interfacing- Pearson T3 Kenneth J Ayala - The 8051 Microcontroller - Thomson - Third Edition. Online Resources: Sl.No Website Link 1 <https://www.electronicshub.org> 2 <https://www.elprocus.com> 3 <https://www.tutorialspoint.com>. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

# Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

## Module 1

1-mark definition: State the meaning of **Compare Microprocessor and Microcontroller**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 2

1-mark definition: State the meaning of **3 Understanding Programming. Explain various addressing modes and types**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 3

1-mark definition: State the meaning of **4 Understanding Interrupts and handling of interrupts. Explain TMOD register and timer operating**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 4

1-mark definition: State the meaning of **4 Understanding 8051. Explain the interfacing of DC motor for speed.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## QUICK REVISION

# Exam checklist

## Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

## Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

## REFERENCE VOCABULARY

# Glossary

### Study glossary

| Term           | Meaning in this lesson                                                   |
|----------------|--------------------------------------------------------------------------|
| Course Outcome | A capability the student should demonstrate after completing the course. |
| Module         | An official syllabus unit with defined topics and hours.                 |
| CIE            | Continuous Internal Evaluation.                                          |
| ESE            | End Semester Examination.                                                |

| Term              | Meaning in this lesson                                                   |
|-------------------|--------------------------------------------------------------------------|
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

## SOURCE-SUPPORTED REFERENCES

### References and web resources

#### References listed in the official PDF

References are included in the official source PDF.

#### Web resources listed in the official PDF

- Sl.No Website Link <https://www.electronicshub.org> <https://www.elprocus.com>  
<https://www.tutorialspoint.com>

## IN-PAGE SEARCH

### Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 5492. Verify institutional updates against the current official curriculum.